

✓ 1. Timesharing is a technique by which CPU time is shared among multiple interactive users at the same time.

a. True      b. False

✓ 2. CPU scheduling is determining which process in memory is given access to the CPU so that it may execute.

a. True      b. False

✓ 3. Memory management is keeping track of how and where programs are loaded into main memory.

a. True      b. False

✓ 4. A process is defined as a running program.

a. True      b. False

✓ 5. The first page in a paged memory system is page 0.

a. True      b. False

✓ 6. A computer can have more than one operating system, but only one OS is in control at any given time.

a. True      b. False

✓ 7. Multiprogramming is the technique of using multiple CPUs to run programs.

a. True

b. False

✓ 8. Multiprogramming does not allow the processor to make use of idle time caused by long-wait interrupt handling.

a. True

b. False

✓ 9. Threads are cheaper to create than processes.

a. True

b. False

✓ 10. User interface is an integral part of an operating system.

a. True

b. False

✗ 11. MS-DOS is a multi-user operating system.

a. True

b. False

✗ 12. Multiprogramming (having more programs in RAM simultaneously) decreases total CPU efficiency (in comparison to Uniprogramming or Batch processing).

a. True

b. False

13. The kernel uses the Process Control Block to keep track of bookkeeping information about processes, such as the program counter.

a. True

b. False

 14. A paging scheme does not suffer from external fragmentation.

a. True

b. False

15. In dual-mode processing, a user process does I/O by accessing the I/O device directly.

a. True

b. False

 16. The operating system is essential in a modern computer system to increase the complexity of user programs.

a. True

b. False

17. The two modes of operation in most computers are Supervisor mode and User mode.

a. True

b. False

 18. The primary role of a Network Manager in an operating system is to manage user interfaces.

a. True

b. False

19. The Memory Manager in an operating system is responsible for executing programs in main memory.

a. True

b. False

20. The Process Manager in an operating system is responsible for managing user and system processes.

a. True

b. False

21. The primary role of the File Manager in an operating system is to create, read, update, and delete files and directories.

a. True

b. False

22. Internal fragmentation appears when the operating system manages fixed memory partitioning.

a. True

b. False

23. Segmentation is one of the efficient memory management techniques.

a. True

b. False

24. In the described sequence of actions, the role of the Device Manager is receiving the electrical impulse from the mouse.

a. True

b. False

25. Hybrid systems combine features of both batch and interactive systems.

a. True

b. False

 26. Multiprogramming (having more programs in RAM simultaneously) decreases total CPU efficiency (in comparison to Uniprogramming or Batch processing).

a. True

b. False

 27. If a deadlock is detected, the OS should always kill the process in the deadlock.

a. True

b. False

 28. Segmentation avoids external memory fragmentation.

a. True

b. False

 29. Threads in a process share the same file descriptors.

a. True

b. False

30. Multiprogramming allows the processor to make use of idle time caused by long-wait interrupt handling.

a. True

b. False

 31. Multithreading is useful for applications that perform a number of essentially independent tasks that do not need to be serialized.

a. True

b. False

 32. Cache memory exploits the principle of locality by providing a small, fast memory between the processor and main memory.

a. True

b. False

 ~~40~~ 40. Multiprogramming (having more programs in RAM simultaneously) decreases total CPU efficiency (in comparison to Uniprogramming or Batch processing).

a. True

b. False

 ~~41~~ 41. Process state indicates the address of the next instruction to be executed for this process.

a. True

b. False

 ~~42~~ 42. The one-to-one model multiplexes many user-level threads to a smaller number of kernel threads.

a. True

b. False

~~43~~ 43. A semaphore is a non-negative integer variable that can be used as a binary signal.

a. True

b. False

~~44.~~ A pipe acts as a conduit allowing two processes to communicate.

a. True

b. False

~~45.~~ Secondary storage in a computer system is used to replace main memory when power is lost.

a. True

b. False

~~46.~~ Virtual memory reduces main memory.

a. True

b. False

~~47.~~ Operating system is a resource allocator that manages hardware resources.

a. True

b. False

13) When main memory is too small to accommodate all jobs, the operating system terminates some jobs to free up space.

A) True

B) False

14) It is not important to distinguish between the execution of operating system code and user-defined code.

A) True

B) False